

1) a) snowflake schema Diagram for city hospital

[Patient-sub-Dim]

- patient-type-id
- patient-type-desc



[Patient-Dim]

- patient-id
- patient-name
- age-group
- patient-type-id

[Doctor-sub-Dim]

- specialty-id
- specialty-name



[Doctor-Dim]

- doctor-id
- doctor-name
- specialty-id

[Hospital-visit-Fact]

- patient-id
- doctor-id
- dept-id
- time-id
- visit-count
- avg-charge

[Department-Dim]

- dept-id
- dept-name
- building

[Time dim]

- time-id
- day
- month-id

[Time sub dim]

- month-id
- month-year

16-11

b) OLAP operations required

* Roll-up on time: Aggregate the time dimension-wise for each year from the lowest level up to the year level.

* Roll-up on patient and doctor: Roll up these dimensions completely to the AU level.

* Keep department: Maintain the department dimension at its primary conceptual level.

c) cuboid count calculation

If a data cube has n dimensions and each dimension i has L_i levels (including the virtual top level AU), the total number of cuboid is calculated using the formula:

$$\text{Total cuboids} = \prod_{i=1}^n L_i$$

$$n=4 \quad L_i=4$$

$$\text{total cuboids} = 4^4 = 256$$

2) Star schema Design

[Time-Dim]

- time-id
- hour, day, week
- month, quarter, year

[Location-Dim]

- location-id
- zone-quadrant
- latitude, longitude

[Smart-city-metris-fact]

- time-id
- location-id
- vehicle-count

- air-quality-index
- noise-level-db

[Transport-line-Dim]

- line-id
- route-name

[Environmental-Assess-Dim]

- asset-id
- sensor-model
- installation-date

Justification of Measures & Dimensions

• Dimensions:

• Time-Dim & Location-Dim are core conformed dimensions shared across all analysis types to correlate traffic spikes with high noise levels or energy grid strains.

• Transport-line-Dim isolates public transit routes to directly calculate efficiency metrics

• Measures:

• vehicle-count & congestion-index satisfy traffic congestion patterns.

• energy-kwh satisfies the utility tracking requirements

• punctuality-rate & passenger-ridership map out public transportation productivity

3) a) Galaxy schema Diagram

[Transaction-Fact]

-customer-id
-branch-id
-time-id
-amount-type-id

⇒ [shared dimensions] ⇐

-customer-dim
-branch-dim
-Time-Dim

[Loan-Fact]

-customer-id
-branch-id
-time-id
-amount-type-id

b) OLAP operations required

1. Roll-up the Time-Dim from its base granularity to the yearly level for both tables.

2. Roll-up the customer-dim and Account-type-dim to the All level.

3. Slice/Group by the Branch-Dim keys.

4. Aggregate measures via the average function: $Avg(\text{amount})$ from transaction-fact and $Avg(\text{loan-amount})$ from loan-fact.

c) cuboid count calculation

Since this is a galaxy schema containing two distinct data cubes that fully share the exact same 4 dimensions, each cube generates an identical maximum space of lattice paths.

$$\circ \text{cuboid for cube 1} = 4^4 = 256$$

However, they share the exact same base, intermediate and apex dimension combinations. The total unique cuboids present in the physical structural metadata layer across the shared warehouse space is 256 cuboids.

4) Manufacturing Quality and Efficiency Data warehouse.

OLAP operation

Action & Target Execution Details

1. Roll-up

Move up the structural hierarchy of the product dimension from Batch $\rightarrow$ type $\rightarrow$ category.

concurrently filter the time dimension where $\text{year} = \text{current-ye}$ and aggregate the defect rate measure across all elements in the factory dimension.

2- Drill-down

Navigate down the data hierarchy.

Filter the Factory dimension down to a specific designated location, then break down the aggregated metrics into lower levels of detail: Department and product type.

3- Slice

perform a selection operation on a single dimension. Fix the supplier dimension constraint to Rating ≥ 4 while viewing production units uniformly across all structural regions.

4- Dice

Define a sub-cube by filtering multiple dimensions simultaneously. Dimensions selected: specific product categories (eg: Automotive, Electronics) AND specific Factory locations (e.g. Chennai).
measures projected: Defect rate and supplier reliability.

5) Retail chain inventory and sales Analytics.

OLAP operation

~~Action & Target~~ Execution Details

1- Roll-up

Aggregate the Inventory turnover measure by moving up the product dimension path from SKU $\rightarrow$ subcategory $\rightarrow$ category, while filtering the time dimension to focus explicitly on the

last quarter across all nodes of the store dimension.

2. Drill-down

transition from a high-level summary to granular detail. Filter the product dimension to category = "Electronics" within a targeted geographical region in the store dimension, then split the total sales revenue down to individual subcategory and individual sku levels.

3. Slice

Apply a strict filter on a single dimension across the entire dataset. Filter the customer dimension strictly to evaluate Gender = 'Male' AND Membership tier = 'Platinum', rendering the isolated total sales revenue across all retail chain stores.

4. Dice

crop a multidimensional warehouse block from the warehouse. specify specific criteria across multiple dimensions: Time, store and product. view the resulting intersecting metrics for total sales revenue and units sold.